

Inverse Freezing in the Ghatak–Sherrington Model

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Abstract

We study the zero-magnetic-field Ghatak–Sherrington model with spin space $\{0, \pm 1\}$ and a positive crystal field. For a nonempty open set of crystal fields, we prove paramagnetism at sufficiently high temperature, replica-symmetry breaking on an intermediate temperature interval, and exact paramagnetism at sufficiently low temperature. Each conclusion holds at every globally maximizing active-spin density. The low-temperature argument compares the interaction energy with the crystal-field penalty and forces all maximizing densities into a dilute window where Panchenko’s exact paramagnetic criterion applies. At intermediate temperature, a Gaussian-channel inequality identifies an exact paramagnetic density edge. Differentiability of the constrained Parisi value then produces a strict pressure improvement beyond that edge. Strict Gaussian Poincaré inequality and an explicit one-step variation exclude every remaining replica-symmetric minimizer. The result establishes an existential form of inverse freezing without determining the transition curves or the number of replica-symmetry-breaking steps.

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1 Setup, phase definitions, and the main theorem

The Ghatak–Sherrington model permits a spin to be inactive, in addition to its two magnetic states. A positive crystal field penalizes the active states, while their interaction can favor a glassy phase. These competing effects make it possible for cooling to restore a paramagnetic phase after an intermediate glassy regime. Our purpose is to establish this reentrance directly from the constrained Parisi formula, with a conclusion that covers every globally maximizing density.

1.1 The model and constrained pressure

Let $(g_{ij})_{1 \leq i < j}$ be independent standard Gaussian random variables. For $\sigma \in \{0, \pm 1\}^N$, define

$$X_N(\sigma) = \frac{1}{\sqrt{N}} \sum_{i < j} g_{ij} \sigma_i \sigma_j, \quad K_N(\sigma) = \sum_{i=1}^N \sigma_i^2. \quad (1.1)$$

For inverse temperature $\beta > 0$ and crystal field $D \in \mathbb{R}$, the partition function is

$$Z_N(\beta, D) = \sum_{\sigma \in \{0, \pm 1\}^N} \exp\{\beta X_N(\sigma) - \beta D K_N(\sigma)\}. \quad (1.2)$$

Thus positive D penalizes active spins. The self-overlap $R_{1,1} = K_N/N$ is the active-spin density, whereas the mutual overlap of two replicas is $R_{1,2} = N^{-1} \sum_i \sigma_i^1 \sigma_i^2$.

Write $Z_{N,k}$ for the sum in (1.2) restricted to $K_N = k$. For $u \in [0, 1]$, let

$$P_{\beta,D}(u) = \lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E} \log Z_{N,k_N}(\beta, D), \quad \frac{k_N}{N} \rightarrow u. \quad (1.3)$$

The limit exists and does not depend on the sequence (k_N) . Panchenko’s theorem gives the constrained value through sufficiently slowly shrinking density windows [10, Theorems 1–2]; Appendix A proves that this value agrees with (1.3). In particular, the endpoint $u = 0$ is the all-zero state, and $u = 1$ is the ordinary zero-field Ising SK model with the additive pressure shift $-\beta D$. The global formula is

$$p(\beta, D) := \lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E} \log Z_N(\beta, D) = \sup_{u \in [0, 1]} P_{\beta,D}(u). \quad (1.4)$$

The covariance convention is $\xi(q) = \beta^2 q^2/2$. Indeed, the covariance of βX_N differs from $N\xi(R_{1,2})$ by a uniformly bounded term, which is permitted in Panchenko’s formulation.

Proposition 1.1 (Upper semicontinuity and attainment). *For fixed (β, D) , the map $u \mapsto P_{\beta,D}(u)$ is upper semicontinuous on $[0, 1]$. Consequently, the supremum in (1.4) is attained.*

Proof. Use the finite-step constrained functional of [10, Theorems 1–2], with physical breakpoints $0 = q_0 \leq q_1 \leq \dots \leq q_k \leq q_{k+1} = u$. Write $q_j = ur_j$, with $0 = r_0 \leq \dots \leq r_{k+1} = 1$. For fixed masses, scaled breakpoints, and multiplier, all Gaussian variances and explicit correction terms are continuous in u . The infimum over this common family equals $P_{\beta,D}(u)$, including the reduced endpoint values. An infimum of continuous functions is upper semicontinuous. Compactness of $[0, 1]$ gives attainment. Appendix A also supplies continuity directly. $\square$

We call any maximizer in (1.4) a *global maximizing density*.

1.2 The constrained Parisi functional

Let $\mathcal{A}$ be the set of distribution functions of probability measures on $[0, 1]$, identified up to Lebesgue-null sets, with metric

$$d_1(\alpha, \tilde{\alpha}) = \int_0^1 |\alpha(r) - \tilde{\alpha}(r)| dr. \quad (1.5)$$

This is a compact metric space: weak compactness of probability measures on $[0, 1]$ gives convergence of distribution functions at continuity points, hence convergence in L^1 by bounded convergence. For $u \in (0, 1)$, $\alpha \in \mathcal{A}$, and $\lambda \in \mathbb{R}$, let $\Phi_{u, \alpha, \lambda}$ solve the backward equation

$$\partial_r \Phi + \frac{\beta^2 u}{2} (\Phi_{xx} + \alpha(r) \Phi_x^2) = 0, \quad 0 \leq r < 1, \quad (1.6)$$

with terminal value

$$\Phi(1, x) = G_\lambda(x) := \log(1 + 2e^{\lambda - \beta D} \cosh x). \quad (1.7)$$

Step distribution functions give the solution by successive Gaussian convolutions and Cole–Hopf transforms; general distribution functions are obtained by d_1 approximation. The PDE is understood almost everywhere in the time variable; for step distribution functions it holds classically between jumps. We use this convention for the Parisi PDE throughout. The needed regularity is recalled in Lemma 3.9.

The fixed-domain constrained functional is

$$\mathcal{P}_{\beta, D}(u; \alpha, \lambda) = -\lambda u + \Phi_{u, \alpha, \lambda}(0, 0) - \frac{\beta^2 u^2}{2} \int_0^1 r \alpha(r) dr. \quad (1.8)$$

Panchenko’s constrained formula becomes

$$P_{\beta, D}(u) = \inf_{\alpha \in \mathcal{A}, \lambda \in \mathbb{R}} \mathcal{P}_{\beta, D}(u; \alpha, \lambda). \quad (1.9)$$

To check the rescaling, put $q = ur$ in the physical overlap interval $[0, u]$. Since $\xi'(q) = \beta^2 q$, the correction is

$$-\frac{1}{2} \int_0^u \alpha(q/u) q d\xi'(q) = -\frac{\beta^2 u^2}{2} \int_0^1 r \alpha(r) dr.$$

The diffusion coefficient transforms to $\beta^2 u/2$, as in (1.6). Formula (1.9) fixes the self-overlap; it is not a replacement of that constraint by a Legendre dual.

A physical order parameter at density u is a probability measure on $[0, u]$. It is *replica symmetric* if it is a single atom δ_q , $0 \leq q \leq u$. The atom δ_0 is represented by $\alpha_{\text{pm}} = 1$ almost everywhere on $(0, 1)$. Its trial value is

$$\mathcal{P}_{\beta, D}(u; \alpha_{\text{pm}}, \lambda) = -\lambda u + \log(1 + 2e^{\lambda - \beta D + \beta^2 u/2}) - \frac{\beta^2 u^2}{4}. \quad (1.10)$$

Minimization over λ gives

$$\text{PM}_{\beta, D}(u) = s(u) - \beta D u + \frac{\beta^2 u^2}{4}, \quad (1.11)$$

$$s(u) = -u \log u - (1 - u) \log(1 - u) + u \log 2, \quad (1.12)$$

$$\lambda_{\text{pm}}(u) = \beta D - \frac{\beta^2 u}{2} + \log \frac{u}{2(1 - u)}. \quad (1.13)$$

We use $0 \log 0 = 0$, so s and PM extend continuously to $[0, 1]$. The admissible trial δ_0 yields $P_{\beta, D}(u) \leq \text{PM}_{\beta, D}(u)$, also at the reduced endpoints.

Definition 1.2 (Constrained and global phases). At density u , the model is *exactly* $q = 0$ *paramagnetic* if $P_{\beta,D}(u) = \text{PM}_{\beta,D}(u)$, equivalently if δ_0 is a constrained Parisi minimizer. It is *replica-symmetry breaking* if no constrained minimizer is a single atom. The global phase is *paramagnetic* if every global maximizing density is exactly $q = 0$ paramagnetic. It is *RSB* if every constrained Parisi minimizer at every global maximizing density is non-replica-symmetric.

Uniqueness is not part of these definitions. For interior densities, existence and uniqueness of the minimizing pair will follow from Lemmas 3.7–3.9. At $u = 1$ we use the usual Ising Parisi problem; at $u = 0$ the only physical order parameter is δ_0 .

1.3 The two crystal-field thresholds

Let $\varphi(a) = (2\pi)^{-1/2}e^{-a^2/2}$ and $\bar{\Phi}(a) = \int_a^\infty \varphi(x) dx$. Define

$$D_{\text{LT}} := \sup_{a \geq 0} \frac{\sqrt{2}\varphi(a)}{\sqrt{\bar{\Phi}(a)}}, \quad \Gamma(u) := \frac{1}{2} + u \log \frac{2(1-u)}{u} \quad (0 < u < 1), \quad (1.14)$$

and put

$$\mathcal{U} := \{u \in (1/3, 1) : D_{\text{LT}} < \Gamma(u)\}. \quad (1.15)$$

Proposition 2.11 proves $D_{\text{LT}} < \Gamma(1/3)$, so $\mathcal{U}$ is nonempty. The lower threshold controls the energy gain of a dilute system; the upper threshold makes the pressure increase at an exact paramagnetic density edge.

Theorem 1.3 (Two-regime inverse freezing). *Fix $u_\star \in \mathcal{U}$ and $D_{\text{LT}} < D < \Gamma(u_\star)$, and set $\beta_\star = 1/u_\star$.*

- (i) *There is $\beta_L < \infty$ such that, for every $\beta \geq \beta_L$, every global maximizing density satisfies $u < \beta^{-3}$ and is exactly $q = 0$ paramagnetic.*
- (ii) *There is $\varepsilon > 0$ such that, for every $\beta > 0$ with $|\beta - \beta_\star| < \varepsilon$, the global phase is replica-symmetry breaking.*

Remark 1.4 (Temperature reentrance). Corollary 2.9 supplies a high-temperature paramagnetic regime, uniformly in D . Together with Theorem 1.3, this gives a paramagnetic regime, an intermediate RSB interval, and a dilute paramagnetic regime as temperature decreases. The theorem does not identify the transition curves, the order of a transition, or whether the intermediate phase is one-step, finite-step, or full RSB.

Related work. The model was introduced to study crystal-field effects in spin glasses [6]. Replica calculations of Crisanti and Leuzzi exhibit inverse freezing in this spin-one setting [5]. Albanese, Barra, and Cirillo derive replica-symmetric and one-step replica-symmetry-breaking pressure formulas by Guerra interpolation for a family of models that includes the Ghatak–Sherrington model [2]. Their formulas use prescribed concentration hypotheses for the overlap, self-overlap, and magnetization [2, Assumptions 1–2 and Theorems 1–2]. The present theorem instead concerns the full constrained Parisi problem and every global maximizing density, without imposing an RS or one-step RSB overlap structure.

Panchenko’s constrained Parisi formula and exact fixed-density replica-symmetric criterion provide our probabilistic foundation [10]. His discussion in Section 2.1 also identifies a global obstruction: maximizing an RS upper bound over the density does not establish that the true pressure exceeds every paramagnetic branch. Our strict improvement of the full constrained

pressure beyond the exact paramagnetic edge addresses this obstruction; the exclusion of all positive-overlap single atoms then gives the RSB conclusion.

The uniqueness theory of Auffinger and Chen [3], and its extension by Hong-Bin Chen to more general spin distributions [4], supply the convexity framework needed to differentiate the constrained value. Gaussian-channel identities [7, 8] enter both the exact-edge calculation and the one-step variation.

Proof strategy. Section 2 bounds the interaction energy at fixed density and compares it with the crystal-field cost. Entropy cannot compensate for their difference at low temperature unless the active density is small; an exact criterion then proves paramagnetism in that window. Section 3 identifies a density u_* at which the paramagnetic criterion holds exactly up to the boundary $\beta_* u_* = 1$. A positive density derivative of the full constrained pressure creates a branch above every paramagnetic branch. Finally, an explicit variation rules out all positive-overlap replica-symmetric minimizers, and a strict pressure gap makes the conclusion persist on an open temperature interval.

2 Low-temperature paramagnetic phase

Fix $D > D_{\text{LT}}$ and write $\delta = D - D_{\text{LT}} > 0$. We first bound the largest interaction energy at a given active density. The crystal-field penalty then leaves an uncompensated cost $\beta\delta u$. At low temperature this cost forces every maximizing density into $u < \beta^{-3}$, where an exact fixed-density criterion identifies the paramagnetic solution.

2.1 Interaction energy and dilution

For $k \in \{0, \dots, N\}$, let

$$\Sigma_{N,k} = \{\sigma \in \{0, \pm 1\}^N : K_N(\sigma) = k\}, \quad M_N(k) = \max_{\sigma \in \Sigma_{N,k}} X_N(\sigma). \quad (2.1)$$

Lemma 2.1 (Energy comparison at fixed density). *Suppose $k_N/N \rightarrow u \in (0, 1]$, and let $a_u \geq 0$ satisfy $u = 2\bar{\Phi}(a_u)$. Then*

$$\limsup_{N \rightarrow \infty} \frac{1}{N} \mathbb{E} M_N(k_N) \leq U(u) := 2\sqrt{u} \varphi(a_u). \quad (2.2)$$

The same conclusion holds at $u = 0$ with $U(0) = 0$.

Proof. Put $u_N = k_N/N$ and compare X_N on Σ_{N,k_N} with

$$Y_N(\sigma) = \sqrt{u_N} \sum_{i=1}^N z_i \sigma_i,$$

where the z_i are independent standard Gaussian variables. For $\sigma, \tau \in \Sigma_{N,k_N}$, write $r = \sum_i \sigma_i \tau_i$ and $s = \sum_i \sigma_i^2 \tau_i^2$. Direct calculation gives

$$\begin{aligned} \mathbb{E}(X_N(\sigma) - X_N(\tau))^2 &= \frac{k_N^2 - k_N - r^2 + s}{N}, \\ \mathbb{E}(Y_N(\sigma) - Y_N(\tau))^2 &= \frac{2k_N(k_N - r)}{N}. \end{aligned}$$

The second quantity minus the first is $((k_N - r)^2 + k_N - s)/N \geq 0$. The Sudakov–Fernique comparison [1] therefore yields

$$\mathbb{E}M_N(k_N) \leq \mathbb{E} \max_{\sigma \in \Sigma_{N,k_N}} Y_N(\sigma) = \sqrt{u_N} \mathbb{E} \sum_{j=1}^{k_N} |z|_{(j)}, \quad (2.3)$$

where $|z|_{(1)} \geq \dots \geq |z|_{(N)}$ are the ordered absolute values. Here and below, Z denotes a standard Gaussian variable. For $u > 0$, the trimmed sum satisfies

$$\frac{1}{N} \sum_{j=1}^{k_N} |z|_{(j)} \longrightarrow \mathbb{E}[|Z| \mathbf{1}_{\{|Z| \geq a_u\}}] = 2\varphi(a_u)$$

in L^1 . One may obtain this by truncating $|Z|$ at a fixed level, using the empirical distribution and its continuous quantiles, and then bounding the discarded tail by $N^{-1} \sum_i |z_i| \mathbf{1}_{\{|z_i| > R\}}$. Its expectation tends to zero as $R \rightarrow \infty$. This proves (2.2).

For the endpoint, Cauchy–Schwarz and Jensen give the finite-volume estimate

$$\frac{1}{N} \mathbb{E}M_N(k) \leq \frac{\sqrt{k/N}}{N} \mathbb{E} \sqrt{k \sum_{i=1}^N z_i^2} \leq \frac{k}{N}. \quad (2.4)$$

Thus the normalized expected maximum tends to zero whenever $k_N/N \rightarrow 0$. $\square$

Lemma 2.2 (Uniformity over density windows). *For every nonempty $I_N \subset \{0, \dots, N\}$, there is a universal constant C such that*

$$\mathbb{E} \max_{k \in I_N} (M_N(k) - \mathbb{E}M_N(k)) \leq C \sqrt{N \log(N+1)}. \quad (2.5)$$

Proof. The Euclidean Lipschitz constant of $M_N(k)$ as a function of the disorder has square

$$\max_{\sigma \in \Sigma_{N,k}} \frac{1}{N} \sum_{i < j} \sigma_i^2 \sigma_j^2 = \frac{k(k-1)}{2N} \leq \frac{N}{2}.$$

Gaussian concentration [9] gives $\mathbb{P}(M_N(k) - \mathbb{E}M_N(k) \geq t) \leq e^{-t^2/N}$ for $t \geq 0$. A union bound over at most $N+1$ shells, followed by integration of $\min\{1, (N+1)e^{-t^2/N}\}$, proves (2.5). $\square$

Lemma 2.3 (Linear energy bound). *The constant D_{LT} is finite, and $U(u) \leq D_{\text{LT}}u$ for every $u \in [0, 1]$.*

Proof. For $u > 0$, the relation $u = 2\bar{\Phi}(a_u)$ gives

$$\frac{U(u)}{u} = \frac{\sqrt{2}\varphi(a_u)}{\sqrt{\bar{\Phi}(a_u)}} \leq D_{\text{LT}}.$$

We use the lower Mills bound

$$\bar{\Phi}(a) \geq \frac{2\varphi(a)}{a + \sqrt{a^2 + 4}}, \quad a \geq 0. \quad (2.6)$$

For completeness, set $m(a) = \bar{\Phi}(a)/\varphi(a)$ and $b(a) = 2/(a + \sqrt{a^2 + 4})$. Then $m' = am - 1$, $b^2 + ab = 1$, and $b' = -b/\sqrt{a^2 + 4}$. Since $b \geq 1/\sqrt{a^2 + 4}$,

$$((m - b)e^{-a^2/2})' = \left(-b^2 + \frac{b}{\sqrt{a^2 + 4}}\right) e^{-a^2/2} \leq 0.$$

Both $me^{-a^2/2}$ and $be^{-a^2/2}$ tend to zero at infinity, so integration gives $m \geq b$. This proves (2.6). Consequently,

$$D_{\text{LT}}^2 \leq \sup_{a \geq 0} \varphi(a)(a + \sqrt{a^2 + 4}) < \infty. \quad (2.7)$$

The endpoint $u = 0$ follows from its definition. $\square$

Lemma 2.4 (Constrained pressure bound). *For every $u \in [0, 1]$,*

$$P_{\beta,D}(u) \leq s(u) + \beta U(u) - \beta Du \leq s(u) - \beta \delta u. \quad (2.8)$$

Proof. At exact density k ,

$$Z_{N,k}(\beta, D) \leq \binom{N}{k} 2^k \exp\{\beta M_N(k) - \beta Dk\}.$$

Take expected logarithms, divide by N , and use Stirling's formula and Lemma 2.1 along $k_N/N \rightarrow u$. Appendix A identifies the limit with $P_{\beta,D}(u)$, including both endpoints. Alternatively, summing over a shrinking density window adds at most $\log(N+1)$ before maximizing over shells; Lemma 2.2 replaces the random maxima by their expectations at an $o(N)$ cost. The same upper bound follows. Lemma 2.3 and $\delta = D - D_{\text{LT}}$ give the second inequality. $\square$

Lemma 2.5 (Entropy bound). *For $u \in (0, 1]$, one has $s(u) \leq u \log(2e/u)$.*

Proof. The inequality $-(1-u) \log(1-u) \leq u$ gives $s(u) \leq -u \log u + u + u \log 2$. $\square$

Proposition 2.6 (Dilution of every low-temperature maximizer). *There is $\beta_1 < \infty$ such that, for every $\beta \geq \beta_1$, every global maximizing density satisfies $u < \beta^{-3}$.*

Proof. Take $\beta_1 \geq 1$. If $u \geq \beta^{-3}$, Lemmas 2.4 and 2.5 imply

$$P_{\beta,D}(u) \leq u(\log(2e\beta^3) - \beta\delta) < 0 \quad (2.9)$$

for all sufficiently large β . But $P_{\beta,D}(0) = 0$, by the all-zero configuration and either (1.3) or the zero-density bound (2.4). Hence no such u maximizes (1.4). $\square$

2.2 Exact paramagnetism in the dilute window

Dilution alone does not identify the Parisi minimizer. We now apply an exact fixed-density criterion, rather than merely a local stability test. For $0 \leq a \leq u$, define

$$L_{\beta,u,a}(z) = 1 - u + ue^{-\beta^2 a/2} \cosh(\beta \sqrt{a} z), \quad (2.10)$$

$$f_{\beta,u}(a) = -\frac{\beta^2 a^2}{4} + \mathbb{E}[L_{\beta,u,a}(Z) \log L_{\beta,u,a}(Z)]. \quad (2.11)$$

Notice that $\mathbb{E}L_{\beta,u,a}(Z) = 1$.

Theorem 2.7 (Panchenko's exact paramagnetic criterion). *For $u \in (0, 1)$,*

$$P_{\beta,D}(u) = \text{PM}_{\beta,D}(u) \iff f_{\beta,u}(a) \leq 0 \text{ for every } a \in [0, u]. \quad (2.12)$$

For each fixed β , the exact $q = 0$ densities form an interval $[0, u_0(\beta)]$. The criterion is independent of D .

This is the spin-one specialization of [10, Theorem 3 and Section 2.1], with the source's crystal-field parameter equal to $-\beta D$. The multiplier (1.13) removes the field from (2.10). We will handle every use of the density-one endpoint by continuity or by the explicit reduced Ising trial, so no finite multiplier is assumed at that endpoint.

Lemma 2.8 (Uniform small-density criterion). *If $u \in [0, 1]$ and*

$$2\beta^2 u^2 e^{\beta^2 u} \leq 1, \quad (2.13)$$

then u is exactly $q = 0$ paramagnetic.

Proof. The case $u = 0$ has already been settled. First let $0 < u < 1$. For $y > 0$, the identity

$$(y - 1)^2 - (y \log y - y + 1) = y(y - 1 - \log y) \geq 0$$

and $\mathbb{E}L = 1$ give $\mathbb{E}[L \log L] \leq \mathbb{E}(L - 1)^2$. Put $A_a = e^{-\beta^2 a/2} \cosh(\beta \sqrt{a} Z)$. Gaussian integration gives $\mathbb{E}A_a = 1$ and

$$\mathbb{E}(A_a - 1)^2 = \cosh(\beta^2 a) - 1 \leq \frac{\beta^4 a^2}{2} e^{\beta^2 a}.$$

Since $L - 1 = u(A_a - 1)$, for $0 \leq a \leq u$ we obtain

$$f_{\beta,u}(a) \leq \beta^2 a^2 \left(-\frac{1}{4} + \frac{\beta^2 u^2}{2} e^{\beta^2 u} \right) \leq 0. \quad (2.14)$$

Apply Theorem 2.7. If (2.13) holds at $u = 1$, it also holds for every $u < 1$. Continuity of $P_{\beta,D}$ from Appendix A, and continuity of $\text{PM}_{\beta,D}$, give equality at $u = 1$ by taking $u \uparrow 1$. $\square$

Corollary 2.9 (Uniform high-temperature paramagnet). *There is $\beta_H > 0$, independent of D , such that every density $u \in [0, 1]$ is exactly $q = 0$ paramagnetic whenever $0 < \beta \leq \beta_H$. Consequently, the global high-temperature phase is paramagnetic.*

Proof. Choose β_H so that $2\beta_H^2 e^{\beta_H^2} \leq 1$. Then $2\beta^2 u^2 e^{\beta^2 u} \leq 2\beta^2 e^{\beta^2} \leq 1$ uniformly in $u \leq 1$, and Lemma 2.8 applies. $\square$

Proposition 2.10 (The dilute window is paramagnetic). *There is $\beta_2 < \infty$ such that every density $0 \leq u \leq \beta^{-3}$ is exactly $q = 0$ paramagnetic for every $\beta \geq \beta_2$.*

Proof. Take $\beta_2 \geq 1$. In the stated window,

$$2\beta^2 u^2 e^{\beta^2 u} \leq 2\beta^{-4} e^{1/\beta} \rightarrow 0.$$

Apply Lemma 2.8. $\square$

Proof of Theorem 1.3(i). Choose $\beta_L \geq \max\{\beta_1, \beta_2\}$. Proposition 2.6 places every maximizing density below β^{-3} , and Proposition 2.10 proves exact paramagnetism throughout that window. The bound $u < \beta^{-3} \rightarrow 0$ also gives dilution. $\square$

2.3 A common crystal-field window

The low-temperature argument requires $D > D_{\text{LT}}$, whereas the intermediate-temperature argument will require $D < \Gamma(u_*)$. The next comparison ensures that both requirements can hold.

Proposition 2.11 (Strict threshold separation). *One has*

$$D_{\text{LT}} < \Gamma(1/3) = \frac{1}{2} + \frac{\log 4}{3}. \quad (2.15)$$

Consequently, the set $\mathcal{U}$ in (1.15) is nonempty and open.

Proof. The logarithmic derivative of $a \mapsto \varphi(a)(a + \sqrt{a^2 + 4})$ is $-a + 1/\sqrt{a^2 + 4}$. Its unique maximum occurs at $a^2 = \sqrt{5} - 2$. Thus (2.7) yields

$$D_{\text{LT}}^2 \leq M := e^{(2-\sqrt{5})/2} \sqrt{\frac{1+\sqrt{5}}{\pi}}.$$

For an entirely analytic strict comparison, use $\sqrt{5} > 20/9$, $\sqrt{5} < 9/4$, and $\pi > 157/50$ to obtain

$$M < e^{-1/9} \sqrt{\frac{325}{314}} < \frac{9}{10} \sqrt{\frac{325}{314}} < \left(\frac{24}{25}\right)^2.$$

The last inequality follows by squaring and comparing integers. On the other hand,

$$\log 2 = 2 \sum_{k=0}^{\infty} \frac{1}{(2k+1)3^{2k+1}} > \frac{56}{81}, \quad \Gamma(1/3) > \frac{467}{486} > \frac{24}{25}.$$

This proves (2.15). Continuity of Γ then gives an interval of admissible $u > 1/3$. $\square$

3 Intermediate-temperature replica-symmetry breaking

Fix $u_* \in \mathcal{U}$, $D_{\text{LT}} < D < \Gamma(u_*)$, and $\beta_* = 1/u_*$. The argument first excludes every globally maximizing paramagnetic branch by a strict pressure comparison. It then excludes replica symmetry with positive overlap at every density. Both exclusions are necessary: beating the $q = 0$ branch alone would not rule out a different single-atom minimizer.

3.1 The exact paramagnetic edge

The function $f_{\beta,u}$ can be differentiated through a scalar Gaussian channel. We record the identities for a general bounded prior, since the positive-overlap variation will require nonsymmetric priors as well.

Lemma 3.1 (Gaussian-channel derivatives). *Let S be a bounded real random variable, independent of Z , and set $Y_t = \sqrt{t}S + Z$. Define*

$$Q(t) = \mathbb{E}[\mathbb{E}(S | Y_t)^2], \quad V_t(y) = \text{Var}(S | Y_t = y), \quad (3.1)$$

$$L_t(y) = \mathbb{E} \exp\{\sqrt{t}yS - tS^2/2\}, \quad \mathcal{D}(t) = \mathbb{E}[L_t(Z) \log L_t(Z)]. \quad (3.2)$$

Then, for $t \geq 0$, with right derivatives at zero,

$$\mathcal{D}'(t) = \frac{1}{2}Q(t), \quad Q'(t) = \mathbb{E}[V_t(Y_t)^2]. \quad (3.3)$$

If S has mean m and variance v , then $\mathcal{D}'(0) = m^2/2$ and $\mathcal{D}''(0) = v^2/2$.

Proof. The function L_t is the density of Y_t relative to standard Gaussian measure. Comparing these two output laws gives

$$\mathcal{D}(t) = \frac{t}{2} \mathbb{E} S^2 - I(S; Y_t), \quad Q(t) = \mathbb{E} S^2 - \text{mmse}(S, t),$$

where I denotes mutual information computed with natural logarithms, in nats, and $\text{mmse}(S, t) = \mathbb{E} \text{Var}(S \mid Y_t)$. The I-MMSE identity [7, Theorem 1] and the MMSE derivative formula [8, Proposition 9] state

$$\frac{d}{dt} I(S; Y_t) = \frac{1}{2} \text{mmse}(S, t), \quad \frac{d}{dt} \text{mmse}(S, t) = -\mathbb{E}[V_t(Y_t)^2].$$

Boundedness of S verifies their moment hypotheses, including the fourth-moment condition for the right derivative at zero. Substitution proves (3.3). At $t = 0$, the posterior equals the prior, giving $Q(0) = m^2$ and $Q'(0) = v^2$. $\square$

Now take the symmetric three-point prior

$$\mathbb{P}(S = 0) = 1 - u, \quad \mathbb{P}(S = 1) = \mathbb{P}(S = -1) = u/2. \quad (3.4)$$

Lemma 3.2 (Posterior-variance bound). *For the prior (3.4), if $1/3 \leq u \leq 1$, then $V_t(y) \leq u$ for all $t \geq 0$ and $y \in \mathbb{R}$. The inequality is strict when $t > 0$ and $u < 1$.*

Proof. Set $r = e^{-t/2}$ and $x = r \cosh(\sqrt{t} y)$, so $0 < r \leq 1$ and $x \geq r$. Bayes' formula gives

$$V_t(y) = \frac{u(1-u)x + u^2 r^2}{(1-u+ux)^2}. \quad (3.5)$$

The desired bound is equivalent to

$$H_{u,r}(x) := (1-u+ux)^2 - (1-u)x - ur^2 \geq 0.$$

For $u \geq 1/2$, this quadratic is increasing on $[0, \infty)$, and

$$H_{u,r}(x) \geq H_{u,r}(r) = (r-1)(u-1)(ru-u+1) \geq 0. \quad (3.6)$$

For $1/3 \leq u < 1/2$, its vertex is $x_0 = (1-u)(1-2u)/(2u^2) \leq 1$. If $x_0 \leq r$, its minimum on $[r, \infty)$ is again at r . If $r < x_0$, then

$$H_{u,r}(x_0) > H_{u,x_0}(x_0) \geq 0,$$

where the last inequality is (3.6) with $r = x_0$. When $r < 1$ and $u < 1$, the boundary value in (3.6) is positive; the second case is already strict. This proves the strictness assertion. $\square$

Lemma 3.3 (Growth of the posterior mean). *For $1/3 \leq u < 1$, the quantity Q_u associated with (3.4) satisfies $Q_u(t) < u^2 t$ for every $t > 0$.*

Proof. For $t > 0$, the posterior variance is positive and strictly less than u . Lemmas 3.1 and 3.2 imply

$$Q'_u(t) = \mathbb{E} V_t(Y_t)^2 < u \mathbb{E} V_t(Y_t) = u(u - Q_u(t)) \leq u^2.$$

Since $Q_u(0) = 0$, integration proves the claim. $\square$

Proposition 3.4 (Exact paramagnetic edge). *Fix $u_\star \in [1/3, 1)$ and set $\beta_\star = 1/u_\star$. At inverse temperature $\beta_\star$, the exact $q = 0$ paramagnetic densities are precisely $[0, u_\star]$.*

Proof. For the prior (3.4), the channel likelihood ratio is $L_t(z) = 1 - u + ue^{-t/2} \cosh(\sqrt{t}z)$. Hence

$$f_{\beta,u}(a) = -\frac{\beta^2 a^2}{4} + \mathcal{D}_u(\beta^2 a), \quad f'_{\beta,u}(a) = \frac{\beta^2}{2} (Q_u(\beta^2 a) - a). \quad (3.7)$$

At $u = u_*$ and $\beta = \beta_*$, Lemma 3.3 gives $Q_{u_*}(\beta_*^2 a) < u_*^2 \beta_*^2 a = a$ for $a > 0$. Therefore

$$f_{\beta_*,u_*}(a) < 0 \quad (0 < a \leq u_*). \quad (3.8)$$

Theorem 2.7 proves exact paramagnetism at u_* , and its interval property gives every smaller density.

For $u_* < v < 1$, Lemma 3.1 gives

$$f''_{\beta_*,v}(0) = \frac{\beta_*^2}{2} (\beta_*^2 v^2 - 1) > 0. \quad (3.9)$$

Since $f(0) = f'(0) = 0$, the exact criterion fails. To check $v = 1$ independently, use the reduced Ising RS trial

$$\mathcal{P}_{\text{RS}}^{\text{SK}}(q) = \mathbb{E} \log(2 \cosh(\beta \sqrt{q} Z)) + \frac{\beta^2}{4} (1 - q)^2. \quad (3.10)$$

Gaussian integration by parts yields

$$\frac{d}{dq} \mathcal{P}_{\text{RS}}^{\text{SK}}(q) = \frac{\beta^2}{2} (q - \mathbb{E} \tanh^2(\beta \sqrt{q} Z)) = \frac{\beta^2}{2} (1 - \beta^2) q + O(q^2) \quad (3.11)$$

as $q \downarrow 0$. The expansion follows from $\tanh^2 x = x^2 + O(x^4)$ and the Gaussian fourth moment. Since $\beta_* > 1$, a small positive q strictly lowers the value at zero. Thus $P_{\beta_*,D}(1) < \text{PM}_{\beta_*,D}(1)$, completing the endpoint exclusion. $\square$

3.2 A strict pressure improvement beyond the edge

The exact edge identifies all paramagnetic densities, but it does not yet show that a different density has higher pressure. We will transfer the positive derivative of the explicit paramagnetic branch to the full constrained variational value. The transfer requires uniqueness, uniform compactness of the multiplier, and differentiability before minimization.

Lemma 3.5 (Positive edge slope). *Suppose $u_* \in [1/3, 1)$, $\beta_* = 1/u_*$, and $D < \Gamma(u_*)$. Then $\partial_u \text{PM}_{\beta_*,D}(u_*) > 0$, and $\text{PM}_{\beta_*,D}$ is strictly increasing on $[0, u_*]$.*

Proof. Differentiating (1.11) gives

$$\partial_u \text{PM}_{\beta,D}(u) = \log \frac{2(1-u)}{u} - \beta D + \frac{\beta^2 u}{2}. \quad (3.12)$$

Thus $u_* \partial_u \text{PM}_{\beta_*,D}(u_*) = \Gamma(u_*) - D > 0$. Also,

$$\partial_{uu} \text{PM}_{\beta_*,D}(u) = \frac{1}{2u_*^2} - \frac{1}{u(1-u)}.$$

If $u_* \leq 1/2$, then for $u \leq u_*$ we have $u(1-u) \leq u_*(1-u_*) \leq 2u_*^2$, since $u_* \geq 1/3$. If $u_* \geq 1/2$, then $u(1-u) \leq 1/4$ and $1/(2u_*^2) \leq 2$. In both cases the second derivative is nonpositive on $(0, u_*]$. The first derivative is therefore at least its positive endpoint value throughout that interval. $\square$

Lemma 3.6 (Identification with Chen's functional). *Fix $u \in (0, 1)$ and $\lambda \in \mathbb{R}$, put $c = \beta^2 u$, and set $\Psi_u(r) = ur$ and $\mu_u(r) = \xi'(\Psi_u(r)) = \beta^2 ur$. With the unnormalized finite measure*

$$\nu_{u,\lambda} = \delta_0 + e^{\lambda - \beta D + c/2}(\delta_{-1} + \delta_1), \quad (3.13)$$

Chen's scalar functional satisfies the exact identity

$$\mathcal{F}_{\nu_{u,\lambda}}(\Psi_u, \alpha) = \mathcal{P}_{\beta,D}(u; \alpha, \lambda) + \lambda u + \frac{\beta^2 u^2}{4}. \quad (3.14)$$

If $\nu_{u,\lambda}$ is normalized to a probability measure, subtract $\log(1 + 2e^{\lambda - \beta D + c/2})$ from the right-hand side.

Proof. Chen's terminal condition tilts the reference measure by $e^{-\mu_u(1)\sigma^2/2}$. Consequently,

$$\log \int e^{x\sigma - c\sigma^2/2} d\nu_{u,\lambda}(\sigma) = \log(1 + 2e^{\lambda - \beta D} \cosh x) = G_\lambda(x).$$

The integral correction is $\frac{1}{2} \int_0^1 \alpha(r) \Psi_u(r) d\mu_u(r) = \frac{1}{2} \beta^2 u^2 \int_0^1 r \alpha(r) dr$. In addition, Chen's definition includes $\theta(u)/2$, where $\theta(u) = u\xi'(u) - \xi(u) = \beta^2 u^2/2$; see [4, equation (1.16), arXiv v2]. These terms give (3.14). Normalizing the measure subtracts its log mass from the terminal value and hence from the solution. All differences in (3.14) are independent of α at fixed (u, λ) ; this is the only invariance used in the strict-convexity application below. $\square$

Lemma 3.7 (Joint strict convexity). *For each fixed $u \in (0, 1)$, the map $(\alpha, \lambda) \mapsto \mathcal{P}_{\beta,D}(u; \alpha, \lambda)$ is jointly strictly convex on $\mathcal{A} \times \mathbb{R}$. In particular, an attained constrained minimum is unique.*

Proof. We use the attained control representation [4, Proposition 3.2, arXiv v2]:

$$\Phi_{u,\alpha,\lambda}(0, 0) = \sup_{v \in \mathcal{V}} \mathbb{E} \left[G_\lambda \left(\beta \sqrt{u} B_1 + \beta^2 u \int_0^1 \alpha(r) v_r dr \right) - \frac{\beta^2 u}{2} \int_0^1 \alpha(r) v_r^2 dr \right]. \quad (3.15)$$

Here B is standard Brownian motion, equipped with its completed, right-continuous natural filtration, and $\mathcal{V}$ consists of controls progressively measurable with respect to this filtration and satisfying $|v_r| \leq 1$. The controlled state is $X_r = \beta \sqrt{u} B_r + \beta^2 u \int_0^r \alpha(s) v_s ds$. The optimizer is the feedback $v_r = \Phi_x(r, X_r)$; bounded spatial derivatives justify the same representation on any subinterval and for the L^1 completion of step cdfs.

Fix λ . The reference measure (3.13) is non-Dirac and supported in $[-1, 1]$, and $\dot{\mu}_u(r) = \beta^2 u > 0$. By Lemma 3.6, Chen's strict-convexity theorem [4, Theorem 1.2] applies to the α dependence.

We make the endpoint convention explicit. The published cdf space permits an atom at zero. The same conclusion follows from the older preprint convention, which imposed $\alpha(0) = 0$, as follows. For two distinct L^1 classes α_0, α_1 , choose $s > 0$ such that their restrictions to $[s, 1]$ differ. Replace each by zero on $[0, s)$ and leave it unchanged on $[s, 1]$. The resulting cdfs satisfy the older convention, and their backward PDE solutions at time s equal those of the original cdfs. The strict joint-in-field inequality in [4, Proposition 3.7, arXiv v2] gives, for $0 < \vartheta < 1$ and any y_0, y_1 ,

$$\Phi_{\alpha_\vartheta}(s, (1 - \vartheta)y_0 + \vartheta y_1) < (1 - \vartheta)\Phi_{\alpha_0}(s, y_0) + \vartheta\Phi_{\alpha_1}(s, y_1),$$

where $\alpha_\vartheta = (1 - \vartheta)\alpha_0 + \vartheta\alpha_1$ and the other parameters are fixed. Use the optimal control for α_ϑ on $[0, s]$ as a trial control for both endpoints. Its terminal fields combine affinely, as do its control costs, so expectation propagates the strict inequality to time zero. This establishes strict convexity even for cdfs with an atom at zero, without a limiting argument that might lose strictness.

To treat distinct multipliers, note that $G_\lambda(x)$ is strictly jointly convex in (x, λ) : its Hessian is the covariance matrix of (σ, σ^2) under weights proportional to $e^{x\sigma + (\lambda - \beta D)\sigma^2}$. All three weights are positive and the points $(-1, 1), (0, 0), (1, 1)$ are not collinear. Take the optimal control for a convex combination of two pairs (α_0, λ_0) and (α_1, λ_1) , and use it at both endpoints. The Brownian path is the same, the terminal fields combine affinely, and the control cost is affine in α . If $\lambda_0 \neq \lambda_1$, strict joint convexity of G gives a strict inequality after expectation. If $\lambda_0 = \lambda_1$ but $\alpha_0 \neq \alpha_1$, strictness was just proved. The remaining terms in (1.8) are affine in (α, λ) , completing the proof. $\square$

Lemma 3.8 (Uniform multiplier coercivity). *Fix $0 < u_- < u_+ < 1$. There is $C < \infty$, depending only on u_-, u_+, β, D , such that, for every $u \in [u_-, u_+]$, $\alpha \in \mathcal{A}$, and $\lambda \in \mathbb{R}$,*

$$\mathcal{P}_{\beta, D}(u; \alpha, \lambda) \geq \begin{cases} (1 - u_+)\lambda - C, & \lambda \geq 0, \\ -u_-\lambda - C, & \lambda < 0. \end{cases} \quad (3.16)$$

Every common finite sublevel of these functionals is therefore contained in a common bounded multiplier interval.

Proof. Since $2 \cosh x \geq 1$, $G_\lambda(x) \geq \lambda - \beta D$; also $G_\lambda(x) \geq 0$. The zero control in (3.15) gives the corresponding lower bounds on $\Phi(0, 0)$. The correction in (1.8) has absolute value at most $\beta^2 u_+^2/4$. Combining these facts with $-\lambda u$ gives both lines of (3.16). $\square$

For the next two statements write $\Phi_{c, \alpha, \lambda}$ for the solution of

$$\Phi_r + \frac{c}{2}(\Phi_{xx} + \alpha \Phi_x^2) = 0, \quad \Phi(1, x) = G_\lambda(x). \quad (3.17)$$

Lemma 3.9 (Uniform PDE stability). *Fix $0 < c_- < c_+ < \infty$ and $L < \infty$. On $[c_-, c_+] \times \mathcal{A} \times [-L, L]$, spatial derivatives of every fixed positive order are uniformly bounded. If $c_n \rightarrow c$, $d_1(\alpha_n, \alpha) \rightarrow 0$, and $\lambda_n \rightarrow \lambda$, then*

$$\Phi_{c_n, \alpha_n, \lambda_n}(0, 0) \longrightarrow \Phi_{c, \alpha, \lambda}(0, 0), \quad \|\partial_x^j \Phi_{c_n, \alpha_n, \lambda_n} - \partial_x^j \Phi_{c, \alpha, \lambda}\|_\infty \longrightarrow 0 \quad (3.18)$$

for $j = 1, 2, 3$, with the supremum over $[0, 1] \times \mathbb{R}$.

Proof. At fixed (c, λ) , the scalar specialization of [4, Corollary 2.12 and Proposition 2.17, arXiv v2] gives the derivative bounds and convergence under d_1 approximation. Its bounded-spin derivative estimates are uniform on the stated compact parameter sets: the normalized prior stays supported in $[-1, 1]$, while its normalizing constant changes only the zeroth-order value. In particular, $|\Phi_x| \leq 1$. The terminal function itself grows like $|x|$ and is not claimed to be bounded.

To cover the full space $\mathcal{A}$ despite the preprint's endpoint convention, approximate α by α_ε , equal to zero on $[0, \varepsilon)$ and to α on $[\varepsilon, 1]$. Then $d_1(\alpha_\varepsilon, \alpha) \leq \varepsilon$. The quoted d_1 stability makes the solution differences uniformly Cauchy on $[0, 1] \times \mathbb{R}$. Uniform bounds on higher spatial derivatives, together with the finite-difference interpolation inequality, make each derivative of fixed order uniformly Cauchy as well. For instance, a difference h with bounded $\partial_x^m h$ satisfies $\|\partial_x^j h\|_\infty \leq C_{j, m} \|h\|_\infty^{1-j/m} \|\partial_x^m h\|_\infty^{j/m}$ for $0 < j < m$. Thus the derivative bounds and stability pass to the d_1 completion, which includes cdfs with an atom at zero.

To handle changes in (c, λ) , first let α be a step function. Write $\Phi_i = \Phi_{c_i, \alpha, \lambda_i}$ and $H = \Phi_1 - \Phi_0$. On each interval of constancy,

$$H_r + \frac{c_1}{2} H_{xx} + \frac{c_1 \alpha}{2} (\Phi_{1, x} + \Phi_{0, x}) H_x = -\frac{c_1 - c_0}{2} (\Phi_{0, xx} + \alpha \Phi_{0, x}^2). \quad (3.19)$$

The terminal difference and its derivatives of every fixed order are bounded by a constant times $|\lambda_1 - \lambda_0|$. The maximum principle applied to H , then successively to its first three derivatives, yields

$$\|H\|_\infty + \sum_{j=1}^3 \|\partial_x^j H\|_\infty \leq C(|c_1 - c_0| + |\lambda_1 - \lambda_0|). \quad (3.20)$$

After differentiation, the leading drift remains bounded; its differentiated coefficients and the differentiated forcing are controlled by finitely many of the spatial bounds above. Zeroth-order terms are handled by Gronwall's inequality. The resulting factor depends on the total time length one, not on the number or positions of the jumps. Step approximation and the quoted d_1 stability pass (3.20) to general α . Finally split a joint parameter change into a change of (c, λ) at fixed α_n , followed by a change of α_n at fixed (c, λ) . This proves (3.18). $\square$

Proposition 3.10 (Uniform differentiability in density). *Fix $0 < u_- < u_+ < 1$ and $L < \infty$. The map $(u, \alpha, \lambda) \mapsto \mathcal{P}_{\beta,D}(u; \alpha, \lambda)$ is continuous on $[u_-, u_+] \times \mathcal{A} \times [-L, L]$, is differentiable in u , and has a jointly continuous density derivative. If X solves*

$$dX_r = \beta^2 u \alpha(r) \Phi_x(r, X_r) dr + \beta \sqrt{u} dB_r, \quad X_0 = 0, \quad (3.21)$$

where $\Phi = \Phi_{u,\alpha,\lambda}$, then

$$\partial_u \mathcal{P}_{\beta,D}(u; \alpha, \lambda) = -\lambda + \frac{\beta^2}{2} \mathbb{E} \int_0^1 (\Phi_{xx} + \alpha \Phi_x^2)(r, X_r) dr - \beta^2 u \int_0^1 r \alpha(r) dr. \quad (3.22)$$

Proof. Put $c = \beta^2 u$. For step α , the solution is a finite composition of Gaussian convolutions and Cole–Hopf transforms and is differentiable in c . Differentiating (3.17), with $w = \partial_c \Phi$, gives

$$w_r + \frac{c}{2} w_{xx} + c \alpha \Phi_x w_x = -\frac{1}{2} (\Phi_{xx} + \alpha \Phi_x^2), \quad w(1, x) = 0.$$

The bounded-source Feynman–Kac formula yields

$$\partial_c \Phi_{c,\alpha,\lambda}(0, 0) = \frac{1}{2} \mathbb{E} \int_0^1 (\Phi_{xx} + \alpha \Phi_x^2)(r, X_r) dr, \quad (3.23)$$

where $dX_r = c \alpha(r) \Phi_x(r, X_r) dr + \sqrt{c} dB_r$.

For general α , call the right-hand side of (3.23) $W(c, \alpha, \lambda)$. We first establish its joint continuity. If $(c_n, \alpha_n, \lambda_n) \rightarrow (c, \alpha, \lambda)$, couple the corresponding diffusions with the same Brownian motion. The drift is uniformly bounded and Lipschitz in space by Lemma 3.9. Subtracting their integral equations, then applying the Brownian maximal estimate and Gronwall's inequality, gives

$$\begin{aligned} \mathbb{E} \sup_{0 \leq r \leq 1} |X_r^{(n)} - X_r|^2 &\leq C(|c_n - c|^2 + d_1(\alpha_n, \alpha)^2) \\ &\quad + C \|\Phi_{c_n, \alpha_n, \lambda_n, x} - \Phi_{c, \alpha, \lambda, x}\|_\infty^2 \longrightarrow 0. \end{aligned} \quad (3.24)$$

Here the square root is Lipschitz on the compact positive c interval, and the time integral of the order-parameter difference is d_1 . In the integral defining W , changes in Φ_{xx} , α , and Φ_x converge by Lemma 3.9; changes of the diffusion location converge by (3.24) and the uniform bound on Φ_{xxx} . Thus W is jointly continuous.

Choose step cdfs $\alpha_n \rightarrow \alpha$ in d_1 . For $0 < c_0 < c_1$, the identity already proved gives

$$\Phi_{c_1, \alpha_n, \lambda}(0, 0) - \Phi_{c_0, \alpha_n, \lambda}(0, 0) = \int_{c_0}^{c_1} W(c, \alpha_n, \lambda) dc.$$

The left side converges by Lemma 3.9. Joint continuity of W , and compactness of the parameter set consisting of the convergent sequence and its limit, give uniform convergence of the integrand on $[c_0, c_1]$. The identity therefore holds for α . The fundamental theorem of calculus proves (3.23) in general. Differentiating the explicit terms in (1.8) and using $\partial_u c = \beta^2$ proves (3.22). We have differentiated the constrained functional itself; the density-dependent comparison constant in (3.14) has not been discarded. $\square$

Lemma 3.11 (Compact envelope lemma). *Let K be a compact metric space, let I be an interval, and let $F : I \times K \rightarrow \mathbb{R}$ be continuous with jointly continuous derivative $\partial_u F$. Set $V(u) = \min_{x \in K} F(u, x)$. If x_0 is the unique minimizer at u_0 , then $V'(u_0) = \partial_u F(u_0, x_0)$. At an endpoint of I , the corresponding one-sided statement holds.*

Proof. Let x_t minimize $F(u_0 + t, \cdot)$. Every subsequential limit of x_t minimizes $F(u_0, \cdot)$, by compactness and continuity, so uniqueness implies $x_t \rightarrow x_0$. For $t > 0$,

$$\begin{aligned} V(u_0 + t) - V(u_0) &\leq F(u_0 + t, x_0) - F(u_0, x_0), \\ V(u_0 + t) - V(u_0) &\geq F(u_0 + t, x_t) - F(u_0, x_t) = \int_{u_0}^{u_0+t} \partial_u F(s, x_t) ds. \end{aligned}$$

Divide by t and use joint continuity to obtain the same limit in both bounds. For $t < 0$, division reverses both inequalities and gives the identical limit. $\square$

Proposition 3.12 (Density-envelope identity). *If $u_0 \in (0, 1)$ is exactly $q = 0$ paramagnetic, then*

$$\partial_{u+} P_{\beta, D}(u_0) = \partial_u \text{PM}_{\beta, D}(u_0). \quad (3.25)$$

Proof. Choose $0 < u_- < u_0 < u_+ < 1$. The trial α_{pm} supplies the uniformly bounded upper value $\text{PM}_{\beta, D}(u)$ on this interval. Lemma 3.8 therefore places all minimizers, and all sufficiently near-minimal competitors, in one common compact multiplier interval $[-L, L]$. Compactness of $\mathcal{A}$ and Proposition 3.10 show that the infimum is attained for every u in the interval. In particular, we may minimize over the fixed compact set $\mathcal{A} \times [-L, L]$.

At u_0 , exact paramagnetism means that $(\alpha_{\text{pm}}, \lambda_{\text{pm}}(u_0))$ is a minimizing pair, and Lemma 3.7 makes it unique. Lemma 3.11 now gives

$$\partial_{u+} P_{\beta, D}(u_0) = \partial_u \mathcal{P}_{\beta, D}(u_0; \alpha_{\text{pm}}, \lambda_{\text{pm}}(u_0)).$$

Differentiate (1.10) while keeping λ fixed:

$$\partial_u \mathcal{P}_{\beta, D}(u; \alpha_{\text{pm}}, \lambda) = -\lambda + \frac{\beta^2}{2} \frac{2e^{\lambda - \beta D + \beta^2 u/2}}{1 + 2e^{\lambda - \beta D + \beta^2 u/2}} - \frac{\beta^2 u}{2}.$$

At $\lambda = \lambda_{\text{pm}}(u)$, the fraction equals u , so this derivative is $-\lambda_{\text{pm}}(u)$. Equations (1.13) and (3.12) identify it with $\partial_u \text{PM}_{\beta, D}(u)$, proving (3.25). $\square$

Proposition 3.13 (Strict gap above every paramagnetic branch). *Under the assumptions of Theorem 1.3, there is $u_1 \in (u_*, 1)$ such that*

$$P_{\beta_*, D}(u_1) > \sup_{\substack{u \in [0, 1]: \\ P_{\beta_*, D}(u) = \text{PM}_{\beta_*, D}(u)}} P_{\beta_*, D}(u). \quad (3.26)$$

Proof. By Proposition 3.4, the exact paramagnetic densities are $[0, u_\star]$. Lemma 3.5 places the largest value among them at $u_\star$. Proposition 3.12 gives

$$\partial_{u+} P_{\beta_\star, D}(u_\star) = \partial_u \text{PM}_{\beta_\star, D}(u_\star) > 0.$$

Hence $P_{\beta_\star, D}(u_\star + t) > P_{\beta_\star, D}(u_\star) = \text{PM}_{\beta_\star, D}(u_\star)$ for all sufficiently small $t > 0$. Choose such a $t < 1 - u_\star$ and put $u_1 = u_\star + t$. $\square$

3.3 Exclusion of positive-overlap replica symmetry

It remains to rule out a single atom δ_q with $q > 0$. At fixed interior density u , the replica-symmetric functional is

$$\begin{aligned} \mathcal{P}_{\text{RS}}(u, q, \lambda) = & -\lambda u - \frac{\beta^2}{4}(u^2 - q^2) \\ & + \mathbb{E} \log \left(1 + 2e^{\lambda - \beta D + \beta^2(u-q)/2} \cosh(\beta \sqrt{q} Z) \right), \quad 0 \leq q \leq u. \end{aligned} \quad (3.27)$$

For $q > 0$, set $A = \lambda - \beta D + \beta^2(u - q)/2$ and define

$$\pi_z(\sigma) = \frac{e^{\beta \sqrt{q} z \sigma + A \sigma^2}}{\sum_{\tau \in \{0, \pm 1\}} e^{\beta \sqrt{q} z \tau + A \tau^2}}. \quad (3.28)$$

Write $\langle F(\sigma) \rangle_z = \sum_{\sigma \in \{0, \pm 1\}} F(\sigma) \pi_z(\sigma)$ for expectation under this one-site law, and set

$$m(z) = \langle \sigma \rangle_z, \quad r(z) = \langle \sigma^2 \rangle_z, \quad v(z) = r(z) - m(z)^2.$$

The letter r here denotes a one-site moment; the argument of the Parisi PDE will always be displayed separately.

Lemma 3.14 (RS stationarity and the boundary). *For $0 < q \leq u < 1$,*

$$\partial_\lambda \mathcal{P}_{\text{RS}} = -u + \mathbb{E} r(Z), \quad \partial_q \mathcal{P}_{\text{RS}} = \frac{\beta^2}{2} (q - \mathbb{E} m(Z)^2). \quad (3.29)$$

An RS constrained minimizer with $q > 0$ must satisfy

$$0 < q < u, \quad u = \mathbb{E} r(Z), \quad q = \mathbb{E} m(Z)^2. \quad (3.30)$$

Proof. The multiplier derivative follows directly. In the q derivative, Gaussian integration by parts turns the field derivative into $\beta^2 \mathbb{E}(r - m^2)/2$. The derivative of the shift contributes $-\beta^2 \mathbb{E} r/2$, and that of the explicit correction contributes $\beta^2 q/2$. This proves (3.29).

A constrained minimizer restricted to the RS family minimizes (3.27); its multiplier is finite by Lemma 3.8. If $q = u$, multiplier stationarity gives

$$\partial_q \mathcal{P}_{\text{RS}}(u, u, \lambda) = \frac{\beta^2}{2} \mathbb{E} v(Z) > 0.$$

The variance is positive because all three one-site probabilities are positive. Moving left lowers the value, excluding $q = u$. Both remaining variables are interior, giving (3.30). $\square$

Lemma 3.15 (Strict Poincaré instability). *Every critical pair satisfying (3.30) obeys $\beta^2 \mathbb{E} v(Z)^2 > 1$.*

Proof. Spin-flip symmetry gives $m(-z) = -m(z)$, so $\mathbb{E}m(Z) = 0$. Differentiation of the conditional mean gives $m'(z) = \beta\sqrt{q}v(z)$. Hence Gaussian Poincaré inequality yields

$$q = \text{Var}(m(Z)) \leq \mathbb{E}m'(Z)^2 = \beta^2 q \mathbb{E}v(Z)^2.$$

Equality in Gaussian Poincaré occurs only for affine functions. For example, the Hermite expansion writes the two sides as $\sum_{n \geq 1} a_n^2$ and $\sum_{n \geq 1} n a_n^2$, so equality removes every coefficient above degree one. Here m is bounded and nonconstant because $q > 0$ and $v(z) > 0$. It is therefore not affine, and the inequality is strict. Divide by q . $\square$

The instability inequality must produce an admissible competitor in the full variational problem. The next computation supplies that competitor and fixes the direction of the variation.

Lemma 3.16 (One-step lowering direction). *Let (q, λ) satisfy (3.30). For $q \leq a \leq u$ and $0 < m_0 \leq 1$, define*

$$\Psi(m_0, a) = -\lambda u - \frac{m_0 \beta^2}{4}(a^2 - q^2) - \frac{\beta^2}{4}(u^2 - a^2) + \frac{1}{m_0} \mathbb{E}_{Z_0} \log \mathbb{E}_{Z_1} X_a(Z_0, Z_1)^{m_0}, \quad (3.31)$$

$$X_a(z_0, z_1) = \sum_{\sigma \in \{0, \pm 1\}} \exp \left\{ \beta \sqrt{q} z_0 \sigma + \beta \sqrt{a - q} z_1 \sigma + \left(\lambda - \beta D + \frac{\beta^2}{2}(u - a) \right) \sigma^2 \right\}, \quad (3.32)$$

where Z_0, Z_1 are independent standard Gaussians. Then $\Psi(1, a) = \mathcal{P}_{\text{RS}}(u, q, \lambda)$. Put $f(a) = \partial_{m_0} \Psi(m_0, a)|_{m_0=1}$. If the RS pair is a constrained Parisi minimizer, then $f(a) \leq 0$ for every $a \in [q, u]$. At a critical pair,

$$f(q) = f'(q) = 0, \quad f''(q) = \frac{\beta^2}{2} (\beta^2 \mathbb{E}v(Z)^2 - 1). \quad (3.33)$$

Proof. The trial has cumulative mass zero below q , mass m_0 on $[q, a]$, and mass one above a , so it is admissible. At $m_0 = 1$, integrating out Z_1 changes the shift $\beta^2(u - a)/2$ to $\beta^2(u - q)/2$; the two correction terms likewise combine to the RS correction. If $f(a) > 0$, then

$$\Psi(m_0, a) = \Psi(1, a) + (m_0 - 1)f(a) + o(1 - m_0) < \Psi(1, a)$$

for $m_0 < 1$ sufficiently close to one. A minimizing RS pair therefore requires $f(a) \leq 0$.

For the derivatives, condition on $Z_0 = z$ and write $t = \beta^2(a - q)$. Let $X_q(z) = \sum_{\sigma} e^{\beta \sqrt{q} z \sigma + A \sigma^2}$. Then

$$X_a(z, Z_1) = X_q(z) L_{\pi_z, t}(Z_1), \quad L_{\pi_z, t}(y) = \mathbb{E}_{\pi_z} e^{\sqrt{t} y \sigma - t \sigma^2 / 2},$$

and $\mathbb{E}_{Z_1} L_{\pi_z, t}(Z_1) = 1$. Differentiating $m_0^{-1} \log \mathbb{E}_{Z_1} X_a^{m_0}$ at one cancels the factor $\log X_q(z)$ and leaves $\mathbb{E}_{Z_1} L_{\pi_z, t} \log L_{\pi_z, t}$. Thus

$$f(a) = -\frac{\beta^2}{4}(a^2 - q^2) + \mathbb{E}_{Z_0} \mathcal{D}_{\pi_{Z_0}}(\beta^2(a - q)). \quad (3.34)$$

Lemma 3.1 gives the first two channel derivatives at zero as $m(z)^2/2$ and $v(z)^2/2$. Because $|\sigma| \leq 1$, these derivatives and the needed domination are uniform over the conditional priors π_z ; see also the conditional MMSE identity in [8, Corollary 2]. Consequently,

$$f'(q) = \frac{\beta^2}{2} (\mathbb{E}m(Z)^2 - q) = 0, \quad f''(q) = -\frac{\beta^2}{2} + \frac{\beta^4}{2} \mathbb{E}v(Z)^2.$$

At $a = q$ the extra Gaussian increment vanishes, so $f(q) = 0$. This proves (3.33). $\square$

Theorem 3.17 (No positive-overlap RS minimizer). *At zero magnetic field and every finite $\beta > 0$, no constrained Parisi minimizer at any density can be replica symmetric with overlap $q > 0$.*

Proof. At $0 < u < 1$, Lemma 3.14 gives $0 < q < u$ and the critical equations. Lemmas 3.15 and 3.16 give $f''(q) > 0$ and $f(q) = f'(q) = 0$. Thus $f(a) > 0$ for $a > q$ sufficiently close to q , contradicting the necessary inequality in Lemma 3.16.

At $u = 1$, use the reduced Ising functional (3.10). At $q = 1$, its derivative is $\beta^2 \mathbb{E} \operatorname{sech}^2(\beta Z)/2 > 0$, so moving left lowers the trial. An interior positive-overlap minimizer would satisfy $q = \mathbb{E} \tanh^2(\beta \sqrt{q} Z)$. Apply the same Poincaré argument with $m(z) = \tanh(\beta \sqrt{q} z)$ and $v(z) = \operatorname{sech}^2(\beta \sqrt{q} z)$, followed by the same one-step construction on the two-state spin space. Every identity in (3.34) remains valid for this bounded prior, yielding the same contradiction. At $u = 0$ positive overlap is impossible. $\square$

Lemma 3.18 (Continuity in inverse temperature). *For fixed D and $u \in [0, 1]$, the map $\beta \mapsto P_{\beta,D}(u)$ is locally Lipschitz on $(0, \infty)$. Moreover,*

$$B(\beta) = \max_{0 \leq u \leq \min(1, 1/\beta)} \operatorname{PM}_{\beta,D}(u) \quad (3.35)$$

is continuous there.

Proof. Restrict the finite-volume partition function to a fixed deterministic nonempty set S_N of configurations, independent of β , and put $F_N(\beta) = N^{-1} \mathbb{E} \log \sum_{\sigma \in S_N} e^{\beta X_N(\sigma) - \beta D K_N(\sigma)}$. Let $\langle \cdot \rangle_\beta$ denote the Gibbs average on S_N with weights proportional to $e^{\beta X_N(\sigma) - \beta D K_N(\sigma)}$. Then $F'_N = N^{-1} \mathbb{E} \langle X_N - D K_N \rangle_\beta$. With $a_{ij}(\sigma) = \sigma_i \sigma_j / \sqrt{N}$, Gaussian integration by parts gives

$$0 \leq \mathbb{E} \langle X_N \rangle_\beta = \beta \sum_{i < j} \mathbb{E} (\langle a_{ij}^2 \rangle_\beta - \langle a_{ij} \rangle_\beta^2) \leq \frac{\beta N}{2}.$$

Since $0 \leq K_N/N \leq 1$, we obtain $|F'_N| \leq \beta/2 + |D|$. Choose any fixed density sequence $k_N/N \rightarrow u$ and take $S_N = \Sigma_{N,k_N}$. The bound passes through (1.3) to any pair of temperatures in a compact interval, proving local Lipschitz continuity. For B , write $u = t \min(1, 1/\beta)$ with $t \in [0, 1]$; the objective is jointly continuous on compact parameter sets, so its maximum over t is continuous. $\square$

3.4 Completion of the intermediate-temperature argument

Proof of Theorem 1.3(ii). At $\beta_\star$, Proposition 3.13 excludes δ_0 from every globally maximizing constrained problem. Theorem 3.17 excludes all remaining single atoms, so the global phase is RSB.

To obtain an open interval, first note that every exact paramagnetic density satisfies $\beta u \leq 1$. For $0 < u < 1$, this follows from $f''_{\beta,u}(0) = \beta^2(\beta^2 u^2 - 1)/2 \leq 0$. At $u = 0$ it is automatic; at $u = 1$, equation (3.11) rules out exact paramagnetism if $\beta > 1$. Thus every exactly paramagnetic branch has pressure at most $B(\beta)$.

At $\beta_\star = 1/u_\star$, the feasible interval in (3.35) is $[0, u_\star]$, and Lemma 3.5 gives $B(\beta_\star) = \operatorname{PM}_{\beta_\star,D}(u_\star)$. Proposition 3.13 supplies $u_1 > u_\star$ and $\eta > 0$ such that

$$P_{\beta_\star,D}(u_1) \geq B(\beta_\star) + 3\eta.$$

By Lemma 3.18, choose $0 < \varepsilon < \beta_\star$ so that

$$P_{\beta,D}(u_1) > B(\beta) + \eta \quad \text{whenever } |\beta - \beta_\star| < \varepsilon.$$

No paramagnetic density can maximize in this interval. Theorem 3.17 holds at every $\beta > 0$, so no positive-overlap single atom can minimize at a maximizing density either. This proves the stated global RSB conclusion. The nonempty crystal-field window was established in Proposition 2.11, completing the theorem. $\square$

A Exact-density constraints and shrinking windows

Panchenko's constrained formula is stated using a sufficiently slowly shrinking self-overlap window. We justify the exact-density formulation used in (1.3), including the endpoints, by a finite-volume comparison. This also avoids choosing temperature-dependent windows when proving continuity in β .

Put $F_{N,k} = N^{-1} \mathbb{E} \log Z_{N,k}(\beta, D)$ and $s_N(k) = N^{-1} \log \left[\binom{N}{k} 2^k \right]$. The following comparison separates the combinatorial entropy from the interaction contribution.

Lemma A.1 (Uniform comparison of density shells). *For $0 \leq k, l \leq N$,*

$$\left| \left(F_{N,l} - s_N(l) + \beta D \frac{l}{N} \right) - \left(F_{N,k} - s_N(k) + \beta D \frac{k}{N} \right) \right| \leq \frac{3}{2} \beta^2 \frac{|l - k|}{N}. \quad (\text{A.1})$$

Proof. Assume $k \leq l$ and set $d = l - k$. Let Ω consist of pairs $\omega = (\tau, A)$, where $\tau \in \Sigma_{N,l}$ and A is a k -element subset of its support. Define $\sigma_i = \tau_i \mathbf{1}_{\{i \in A\}}$. On this common finite index set consider the centered Gaussian processes $H_1(\omega) = X_N(\tau)$ and $H_0(\omega) = X_N(\sigma)$, with covariance kernels C_1, C_0 . At a single index,

$$0 \leq (C_1 - C_0)(\omega, \omega) = \frac{l(l-1) - k(k-1)}{2N} \leq d.$$

For two indices ω, ω' , a covariance summand can change only if its edge meets a coordinate removed from τ or τ' . There are at most $2d$ such coordinates and at most $2dN$ incident edges. Restriction can turn a coefficient to zero but cannot flip its sign, so each changed summand has absolute value at most $1/N$. Hence $|(C_1 - C_0)(\omega, \omega')| \leq 2d$.

Interpolate independent copies by $H_t = \sqrt{t}H_1 + \sqrt{1-t}H_0$. Conditional on the Gaussian disorder, let ω, ω' be independent samples from the Gibbs measure on Ω with weights proportional to $e^{\beta H_t(\omega)}$, and write $\langle \cdot \rangle_t$ for their joint average. Gaussian integration by parts gives

$$\frac{d}{dt} \mathbb{E} \log \sum_{\omega \in \Omega} e^{\beta H_t(\omega)} = \frac{\beta^2}{2} \mathbb{E} \langle (C_1 - C_0)(\omega, \omega) - (C_1 - C_0)(\omega, \omega') \rangle_t,$$

whose absolute value is at most $3\beta^2 d/2$. At $t = 1$, each l -spin configuration appears $\binom{l}{k}$ times; at $t = 0$, each k -spin configuration appears $\binom{N-k}{l-k} 2^{l-k}$ times. The identity $\binom{N}{k} \binom{N-k}{l-k} = \binom{N}{l} \binom{l}{k}$ converts these multiplicities exactly into $N(s_N(l) - s_N(k))$. Integrating, dividing by N , and restoring the deterministic crystal-field terms proves (A.1). $\square$

Proposition A.2 (Equivalence of constraints and continuity). *For every $u \in [0, 1]$, all sequences $k_N/N \rightarrow u$ have the same limit in (1.3), equal to Panchenko's constrained Parisi value. The same limit holds for every shrinking window $|K_N/N - u| \leq \varepsilon_N$, with $\varepsilon_N \rightarrow 0$, provided the window is eventually nonempty. Moreover, for $u, v \in [0, 1]$,*

$$|[P_{\beta,D}(v) - s(v) + \beta Dv] - [P_{\beta,D}(u) - s(u) + \beta Du]| \leq \frac{3}{2} \beta^2 |v - u|. \quad (\text{A.2})$$

In particular, $P_{\beta,D}$ is continuous on $[0, 1]$.

Proof. The binomial entropy bounds give $|s_N(k) - s(k/N)| \leq \log(N+1)/N$ uniformly in k . If ω_s denotes the modulus of continuity of s on $[0, 1]$, Lemma A.1 implies

$$|F_{N,l} - F_{N,k}| \leq \omega_s\left(\frac{|l-k|}{N}\right) + \left(\beta|D| + \frac{3}{2}\beta^2\right) \frac{|l-k|}{N} + \frac{2\log(N+1)}{N}. \quad (\text{A.3})$$

The gradient of $\log Z_{N,k}$ is a Gibbs average of vectors $(\beta\sigma_i\sigma_j/\sqrt{N})_{i<j}$, whose squared norms are at most $\beta^2 N/2$. Gaussian concentration and the same integrated union bound as in Lemma 2.2 therefore give, for any nonempty set $I \subset \{0, \dots, N\}$,

$$0 \leq \frac{1}{N} \mathbb{E} \log \sum_{k \in I} Z_{N,k} - \max_{k \in I} F_{N,k} \leq \frac{\log(N+1)}{N} + C\beta \sqrt{\frac{\log(N+1)}{N}}. \quad (\text{A.4})$$

Take $I_N = \{j : |j/N - u| \leq \varepsilon_N\}$. For any sequence $k_N/N \rightarrow u$ and every $j \in I_N$,

$$\frac{|j - k_N|}{N} \leq \varepsilon_N + |k_N/N - u| \rightarrow 0.$$

Equations (A.3) and (A.4) show that the expected normalized log partition function in this window differs from F_{N,k_N} by $o(1)$. Choose first a sufficiently slow window for which Panchenko's theorem supplies convergence [10, Theorems 1–2 and Appendix B]. Every exact-density sequence therefore converges to the same value. Applying the comparison once more gives every eventually nonempty shrinking window, without a rate restriction. Finally, pass (A.1) to exact-density sequences approaching u and v to obtain (A.2). $\square$

At density zero, choosing $k_N = 0$ gives $P_{\beta,D}(0) = 0$. At density one, choosing $k_N = N$ gives

$$Z_{N,N}(\beta, D) = e^{-\beta D N} Z_N^{\text{SK}}(\beta), \quad P_{\beta,D}(1) = p_{\text{SK}}(\beta) - \beta D.$$

For its variational interpretation, note that

$$G_\lambda(x) - \lambda + \beta D = \log(2 \cosh x + e^{-\lambda + \beta D}) \rightarrow \log(2 \cosh x)$$

uniformly in x as $\lambda \rightarrow +\infty$, with error at most $e^{-\lambda + \beta D}/2$. The Parisi recursion preserves constants and is stable in the supremum norm, so the density-one functional, after cancellation of $-\lambda$, converges uniformly in α to the Ising Parisi functional minus βD . Thus endpoint order-parameter minimizers refer to this reduced problem; no finite minimizing multiplier is asserted at $u = 1$.

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